

Pressure and temperature dependent UNIQUAC model for methanol - water mixtures

Adina Werner^{a,*}, Jongmin Kim^{b,*}, Fabian Mauss^a

^a Chair of Thermodynamics/Thermal Process Engineering, Brandenburg University of Technology Cottbus-Senftenberg, Siemens-Halske-Ring 8, 03046 Cottbus, Germany

^b LOGE Deutschland GmbH, Querstraße 48, 03044 Cottbus, Germany

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ABSTRACT

A pressure dependency is included in a quadratic temperature dependent binary interaction parameter of the UNIQUAC model. The obtained activity coefficients for methanol-water mixtures are compared with only temperature dependent UNIQUAC and UNIFAC, and with calculated activity coefficients based on experimental data between 298.15 - 373.15 K and 0.1519 - 1.01325 bar produced with vapor-liquid equilibrium calculations and Wilson method. This model exhibits an overall good agreement. The predicted activity coefficients are more adaptable than those from models without pressure dependence, indicating potential for further improvement.

1. Introduction

Prediction of the real equilibrium behaviour of mixtures remains a challenge. Experimental measurements of even binary mixtures at different compositions, temperatures and pressures are very costly and time-consuming. The execution for numerous existing binary mixtures and even more mixtures with more than two components is not feasible. The accurate prediction of the different equilibria is needed. The work focuses on the vapor-liquid equilibrium of the methanol-water mixture under different temperature and pressure conditions.

The activity coefficients are served to describe real liquid phase and the fugacity coefficients to describe the real vapor phase. In principle, activity coefficients depend on temperature and pressure. As the temperature dependency is much stronger than the pressure dependency, the pressure dependency is often neglected. Models that predict the activity coefficients with the help of the excess free enthalpy are very common. They are called g^E -models. The UNIQUAC and UNIFAC models are well-known [1].

The first version of the UNIQUAC model was suggest by Abrams and Prausnitz [2]. The binary interaction parameter, which represents the interaction energy between two different molecules in a mixture, was neither temperature nor pressure dependent even when the authors figured out a temperature dependency. The research was performed to include the temperature dependency in g^E -models. Several suggestions were made. Skjold-Jørgensen et al. [3] assumed the mesh coordinate number to be temperature dependent as number of nearest neighbors of

a molecule will change with temperature. The method will influence the residual and the combinatorial part at the same time. The focus in this work is on methods which include the temperature dependency only in the binary interaction parameter, which is only part of the residual term in UNIQUAC but also in UNIFAC.

Demirel and Geceğörmüş [4] compared the maximum heat of mixing calculated using UNIQUAC to experimental values. They concluded that the temperature dependent binary interaction parameters are crucial. Unfortunately, their suggested approach had problems, especially for mixtures containing alcohols. Anderson and Prausnitz [5] published a revised version of the original UNIQUAC equation by replacing the old area parameters with new values for alcohols and water. Such parameters, which consider the intermolecular attraction induced by the OH group, are smaller than before. Based on these methods, Demirel et al. [6] updated their parameter which should now give better results for polar mixtures, such as alcohols and water.

Nagata and Gmehling [7] suggested a quadratic temperature dependency of the binary interaction parameters of the UNIQUAC model to calculate the excess free enthalpy. Nagata et al. [8] extended the quadratic temperature dependency of the binary interaction parameters to calculate activity coefficients instead of the excess free enthalpy. The results showed a better agreement with experimental data compared to the results of Demirel and Geceğörmüş [4].

The UNIQUAC method is mostly used under atmospheric pressures where the pressure dependency can be neglected. Only few publications are available which consider the temperature dependency as well as the pressure dependency of the binary interaction parameter. Islam et al. [9]

* Corresponding authors.

E-mail addresses: adina.werner@b-tu.de (A. Werner), jongmin.kim@logesoft.com (J. Kim).

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List of Symbols			
A_{ij}	- Parameter in the calculation approaches of Δu_{ji} in UNIQUAC	T	K Temperature
a	- Parameter in the polar term of the Tsonopoulos method	TD	Temperature dependent
a_{mn}	K Group interaction parameter between the groups m and n	TPD	Temperature-pressure dependent
B	m^3/mol Second virial coefficient	Δu_{ji}	K Binary interaction parameter between molecule i and j in UNIQUAC
B_{ii}	m^3/mol Second virial coefficient of pure component i	UNIFAC	UNIQUAC Functional-Group Activity Coefficients
B_{ij}	- Parameter in the calculation approaches of Δu_{ji} in UNIQUAC	UNIFAC Do	UNIFAC Dortmund
B_{ij}	m^3/mol Cross virial coefficient	UNIQUAC	Universal quasi-chemical
B_M	m^3/mol Second virial coefficient of the mixture	V	m^3 Volume
b	- Parameter in the polar term of the Tsonopoulos method	V1, V2	Version 1, Version 2
$b^{(0)}, b^{(1)}, b^{(2)}$	- Dimensionless terms in the Tsonopoulos method	$\bar{v}$	m^3/mol Partial molar volume
b_{ij}	- Parameter in UNIQUAC Islam	$\bar{x}_i$	- Mole fraction of component i in liquid phase
b_{mn}	- Group interaction parameter between the groups m and n	$\bar{y}_i$	- Mole fraction of component i in vapor phase
C_{ij}	- Parameter in the calculation approaches of Δu_{ji} in UNIQUAC	z_{RA}	- Constant of the modified Rackett equation
c_{ij}	- Parameter in UNIQUAC Islam	<i>Greek letter</i>	
c_{mn}	K^{-1} Group interaction parameter between the groups m and n	γ	- Activity coefficient
D	% Deviation in area test	ϑ	$^{\circ}C$ Temperature
F	- Objective function	$\Lambda_{12}, \Lambda_{21}$	- Wilson parameter
K_{ij}	- Parameter in UNIQUAC Islam	τ_{ji}	- Binary interaction parameter between molecule i and j in UNIQUAC
N	Number of experimental data points	φ_i	- Fugacity coefficient of component i in the mixture
n	mol Number of moles	φ_i^S	- Fugacity coefficient of component i in the pure vapor phase
n_i	- Number of data sets	ψ_{mn}	- Group interaction parameter
n_k	- Number of properties	ω	- Acentric factor
NIST	National Institute of Standards and Technology	<i>Superscript</i>	
NRTL	Non Random Two Liquid	L	Liquid
p	Pa Pressure	<i>Subscript</i>	
p_i^S	Pa Saturation vapor pressure of component i	crit	Critical
Q_k	- Group area parameter	exp	Experimental
$\bar{R}$	$J/(molK)$ Universal gas constant	i, j	Variable component
R_k	- Group volume parameter	R	Reduced
		tot	Total

compared two-parameter models such as UNIQUAC and three-parameter models such as NRTL, which has no temperature dependency. Since they observed that three-parameter models better reproduce the increase in the activity coefficients around small mole fractions, Islam and Kabadi [10] included first the quadratic temperature dependency to the NRTL interaction parameters. Later, Islam and Carlson [11] designed the binary interaction parameters for the UNIQUAC method that can cover a wide range of temperature and also pressure. Accurate correlation of binary VLE data is crucial, as inadequate correlation can lead to strong deviations between experimental data sets and model predictions. Activity coefficients model such as UNIQUAC often results in poor VLE correlations for many binary systems. In addition, an approach of a binary interaction parameter that can consider the temperature and pressure dependencies is currently limited. Further investigations necessary to develop a model that is able to predict the temperature dependency, which is needed for many mixtures, but also the pressure dependency, which is often neglected but should be considered to capture reliable the vapor-liquid equilibrium predictions.

The binary mixture of methanol-water is commonly used to evaluate the performance of predictive thermodynamic models. Methanol is one of the most important chemicals. It can be utilized as a solvent, chemical or fuel. This can be produced via methanol synthesis out of syngas, which consist of hydrogen, carbon monoxide and carbon dioxide. The technology is becoming more popular as it is consuming carbon dioxide. A byproduct of the methanol synthesis is water. Separating methanol

from water is essential to obtain the methanol with high purity [12–14].

In this work, a suggestion of a temperature and pressure dependent binary interaction parameter of the UNIQUAC methods are made to calculate the activity coefficients of methanol-water mixtures at temperatures between 298.15 K to 373.15 K and pressures between 0.1519 bar to 1.01325 bar. The results are compared with the temperature dependent UNIQUAC and UNIFAC and also with calculated activity coefficient based on experimental vapor-liquid equilibrium data.

2. Simulation approaches

The models were implemented in a code written in Fortran.

2.1. UNIFAC dortmund

The UNIQUAC Functional-group Activity Coefficients Dortmund (UNIFAC Do) model [15] is a further development of the UNIFAC model [16] which is based on the group-contribution concept. The concept handles mixtures as a mixture of functional groups instead of a mixture of pure components. The calculation of the activity coefficient is split into a combinatorial and a residual part. The combinatorial part recognises the structural aspects of the groups in an area fraction and segment fraction which are based on the group area parameter (Q_k) and the group volume parameter (R_k) which can be found in a literature [17]. The residual part contains the group interaction parameter ψ_{mn} which was extended by a further temperature dependency compared to

the original UNIFAC model:

$$\psi_{mn} = \exp \left\{ - \frac{(a_{mn} + b_{mn}T + c_{mn}T^2)}{T} \right\} \quad (1)$$

The group interaction parameters a_{mn} , b_{mn} and c_{mn} were obtained by an optimization process using experimental data of activity coefficients at infinity dilution, vapor-liquid equilibria and excess enthalpies at the same time. The values of those parameters are reported in the Dortmund Data Bank [17,15]

For constant temperature simulations, the activity coefficients can be directly calculated with the UNIFAC Do method. The equilibrium pressure p for each composition is estimated for the methanol (1) – water (2) mixtures with:

$$p = \gamma_1 \bar{x}_1 p_1^s + \gamma_2 \bar{x}_2 p_2^s \quad (2)$$

with γ_i as activity coefficients, $\bar{x}_i$ as liquid mole fractions and p_i^s as saturation vapor pressures. The saturation vapor pressures are computed with the following Antoine equations:

$$\log(p_1^s) = 7.19736 - \frac{1574.99}{T - 34.29} \quad (3)$$

$$\log(p_2^s) = 7.11572 - \frac{1684.123}{T - 43.568} \quad (4)$$

The pressures p_i^s are expressed in the unit of kPa and the temperature T the unit of K . Eq. (3) and Eq. (4) are from Gao et al. [18] and Kurihara et al. [19], respectively. Their saturation pressures are in good agreement with the used experimental data. With the activity coefficients the vapor mole fractions are calculated as follows:

$$\bar{y}_i = \frac{\gamma_i \bar{x}_i p_i^s}{p} \quad (5)$$

To carry out the UNIFAC Do simulations at constant pressure, one needs to find the equilibrium temperature of each mixture through an iteration loop. The activity coefficients are calculated from an initial guess temperature. The pressure is obtained by Eq. (2) and then it is compared with the set pressure. If they are not equal, temperature is slightly adjusted, and the loop restarts with the calculation of the activity coefficients. The system temperature is determined when the calculated and set pressures are equal. Afterwards, the vapor mole fractions are obtained using Eq. (5).

2.2. UNIQUAC

Abrams and Prausnitz [2] developed the universal quasi-chemical (UNIQUAC) model, which is based on the local-composition concept from Wilson [20]. It was extended to molecules of different sizes and shapes by incorporating a volume and an area parameter for each pure component. Similarly to the UNIFAC model, the activity coefficients are calculated from a combinatorial and a residual part. The residual part contains the temperature dependency including in the binary interaction parameter τ_{ji} between molecule i and j :

$$\tau_{ji} = \exp \left\{ - \Delta u_{ji} / T \right\} \quad (6)$$

In case of water–methanol mixtures, the binary interaction parameters Δu_{ji} in K were taken from Gmehling and Kolbe [21] with values of $\Delta u_{12} = -165.3 K$ and $\Delta u_{21} = 254.7 K$. No further pressure or temperature specifications were given. Using these parameters over a wide pressure and temperature range can induce uncertainties [2,21]

The simulations for constant pressure and constant temperature are conducted in the same way as with UNIFAC Do. The difference is that UNIFAC Do is replaced by UNIQUAC. Furthermore, a different schema is needed when the binary interaction parameters depend on pressure (see 3 UNIQUAC: Temperature and Pressure Dependent Binary Interaction Parameters). The constant pressure simulations can be performed as

before, including the pressure dependency of the activity coefficient in the existing iteration loop. The constant temperature simulations are extended by an iteration loop. Since a pressure is needed for the calculation of the activity coefficients, a starting pressure is guessed. With the guessed pressure calculated activity coefficients, one can solve Eq. (2) to obtain a new pressure. If the pressures are different, the new pressure is served to calculate the activity coefficients for the next iteration. The loop is repeated until both pressures are equal.

3. UNIQUAC: temperature and pressure dependent binary interaction parameters

3.1. UNIQUAC TD

Several studies proposed involving the temperature dependency in the binary interaction parameters. Nagata and Gmehling [7] suggested a quadratic temperature dependency of the binary interaction parameters of the UNIQUAC model:

$$\Delta u_{ij} = A_{ij} + B_{ij}T + C_{ij}T^2 \quad (7)$$

The parameters A_{ij} , B_{ij} and C_{ij} were fitted for experimental data. As the model includes only temperature dependency (TD), we refer to the model as UNIQUAC TD.

We obtain the optimal parameters by minimizing an objective function F based on the averaged relative deviation. It is defined as:

$$F = \frac{1}{n_i n_k} \sum_k \sum_i \frac{\gamma_{i,k}(exp) - \gamma_{i,k}(calc)}{\gamma_{i,k}(exp)} \quad (8)$$

where n_i and n_k indicate the number of data sets and the number of properties, respectively [11]. In our system, n_k is two for the activity coefficients for methanol and water. $\gamma_{i,k}(calc)$ is obtained using the UNIQUAC method with experimental $\bar{x}_i$, T and p data. $\gamma_{i,k}(exp)$ are the experimental activity coefficients (see 4.1 Experimental Data). The optimization is performed using the Nelder-Mead method. For the optimization Python is used.

3.2. UNIQUAC Islam

Islam and Carlson [11] designed the binary interaction parameters for the UNIQUAC method that can include not only the temperature dependency but also a pressure dependency:

$$\Delta u_{ij} = A_{ij} + B_{ij}T_R^{b_{ij}} + C_{ij}p_R^{c_{ij}} + K_{ij}T_R p_R \quad (9)$$

In order to make the model suitable for a large temperature and pressure range, the reduced temperature T_R and reduced pressure p_R were employed instead of using temperature and pressure. No calculation equation for the reduced values was given by Islam and Carlson [11]. In this work the reduced values are obtained over:

$$T_R = T / \sum_i \bar{x}_i T_{i,crit} \quad (10)$$

$$p_R = p / \sum_i \bar{x}_i p_{i,crit} \quad (11)$$

All 12 parameter of a binary mixture (A_{ij} , A_{ji} , B_{ij} , B_{ji} , b_{ij} , b_{ji} , C_{ij} , C_{ji} , c_{ij} , c_{ji} , K_{ij} and K_{ji}) are estimated by minimizing the objective function (Eq. (8)). In this work, we refer to the model as UNIQUAC Islam [11].

3.3. UNIQUAC TPD

The quadratic temperature dependency of the binary interaction parameter was suggested for the UNIFAC Dortmund as well as for the UNIQUAC method. In this paper, we devise a quadratic dependency

model that also includes the pressure dependency in the following manner:

$$\Delta u_{ij} = A_{ij} + B_{ij}Tp + C_{ij}(Tp)^2 \quad (12)$$

The binary interaction parameter Δu_{ij} and the temperature are in the unit of K , the unit of pressure is *bar*. Temperature and pressure are coupled within the same term. For a pressure of one bar, Eq. (12) will become the pure quadratic temperature dependent equation again. The equation combines the temperature and the pressure dependency and is termed Temperature-Pressure-Dependent (TPD) model. The three parameters A_{ij} , B_{ij} and C_{ij} are obtained using the same optimization approach employed in the UNIQUAC TD model.

4. Experimental data and comparable calculated activity coefficients

4.1. Experimental data

To validate the simulated activity coefficients, experimental data are needed. As the activity coefficients cannot be measured directly, measurements of equilibrium data are required. Different authors examined different measurements and used different methods to calculate out of them the activity coefficients. Three methods were investigated more closely to transfer them to other experimental equilibrium data sets and to compare their performance under different conditions and also with the simulated UNIFAC Do and UNIQUAC activity coefficients.

The calculation of the activity coefficients needs the mole fractions of the two components i in the vapor phase ($\bar{y}_i$) and in the liquid phase ($\bar{x}_i$), the equilibrium pressure p and the equilibrium temperature T . Equilibrium data can be obtained either at constant temperature or at constant pressure. The following paragraphs give an overview of the used data sets. Some data sets were published along with calculated activity coefficients.

The following references were used as data sources of vapor-liquid equilibrium measurements at constant temperature. Poling et al. [22] provided equilibrium data (p , $\bar{x}_i$, $\bar{y}_i$) at constant temperatures of 298.15 K, 323.15 K and 373.15 K. Yao et al. [23] measured equilibrium data (p , $\bar{x}_i$, $\bar{y}_i$) at 318.15 K. McGlashan and Williamson [24] performed equilibrium measurements (p , $\bar{x}_i$, $\bar{y}_i$) at 308.15 K, 323.15 K and 338.15 K. Christensen [25] and Kurihara et al. [19] employed their obtained equilibrium measurements to calculate activity coefficients out of them. Christensen [25] chose five temperature points between 60 °C and 100 °C with very low concentrated methanol and measured p , $\bar{x}_i$ and $\bar{y}_i$. The calculation of the infinite dilution activity coefficient of methanol were carried out similarly to the method of Kurihara et al. [19]. Kurihara et al. [19] conducted measurements for $\bar{x}_i$ and $\bar{y}_i$ at 323.15 K, 328.15 K and 333.15 K. The calculation method of the activity coefficients will be explained in the Section 4.3 Equilibrium Method based on Kurihara et al..

Vapor-liquid equilibrium data at all constant pressure were taken from references which also calculated the activity coefficients for the methanol-water system. Soujanya et al. [26] obtained T and $\bar{x}_i$ at six constant system pressures between 0.1519 bar and 0.953 bar experimentally. The vapor mole fractions were calculated based on the activity coefficients estimated by the Wilson method. The method will be presented in detail in the Section 4.4 Wilson Method based on Soujanya et al.. Gao et al. [18] investigated the vapor-liquid equilibrium at 1.01325 bar, measuring T and $\bar{x}_i$. The provided activity coefficients are based on the NRTL model. The method requires the acquisition of the three model parameters that were optimized via the minimization of an objective function including calculated and experimental mole fractions of the liquid phase. Khalfaoui et al. [27] measured vapor-liquid equilibrium data, T , $\bar{x}_i$ and $\bar{y}_i$, at constant pressures around one bar. Out of the equilibrium data the activity coefficients were calculated (see Section 4.2 Ideal Equilibrium Method based on Khalfaoui et al.). Kato et al.

[28] investigated the vapor-liquid equilibrium also at 1.01325 bar. The measured T , $\bar{x}_i$ and $\bar{y}_i$ equilibrium data were used to calculate the activity coefficients with the Wilson method. As its optimization approach is not clearly described, the Wilson method from Soujanya et al. [26] was applied to other data sets.

4.2. Ideal equilibrium method based on Khalfaoui et al

Khalifaoui et al. [27] calculated the activity coefficients using the following vapor-liquid equilibrium condition:

$$\bar{y}_i p = \gamma_i \bar{x}_i p_i^s \quad (13)$$

The gas phase is handled as an ideal gas. Because of it, we refer to this method as Ideal Equilibrium in this paper. The saturation vapor pressures for methanol (1) and water (2) are calculated with the Antoine equations:

$$\log(p_1^s / \text{bar} \cdot 0.00133322) = 8.07246 - \frac{1574.99}{\vartheta / ^\circ\text{C} + 238.86} \quad (14)$$

$$\log(p_2^s / \text{bar} \cdot 0.00133322) = 7.96681 - \frac{1668.21}{\vartheta / ^\circ\text{C} + 228.0} \quad (15)$$

[27]

The method is adopted to all other constant pressure and constant temperature data sets using the provided $\bar{x}_i$, $\bar{y}_i$, p and T data.

4.3. Equilibrium method based on Kurihara et al

Kurihara et al. [19] calculated the activity coefficients with the following vapor-liquid equilibrium equation:

$$\varphi_i \bar{y}_i p = \gamma_i \bar{x}_i p_i^s \varphi_i^s \exp \left\{ \frac{\bar{v}_i^L (p - p_i^s)}{\bar{R}T} \right\} \quad (16)$$

with φ_i and φ_i^s being fugacity coefficient of component i in the mixture and in the pure vapor phase, $\bar{v}_i^L$ represents the partial molar volumes of component i in the liquid phase. p_i^s is the saturation vapor pressure of component i and $\bar{R}$ is the universal gas constant. Compared to the Ideal Equilibrium method here the gas phase is not treated as an ideal gas. Therefore, the method is called as Equilibrium method [19].

The p_i^s for methanol (1) and water (2) are calculated with the Antoine equation:

$$\log(p_1^s / \text{kPa}) = 7.25325 - \frac{1608.512}{T/K - 31.143} \quad (17)$$

$$\log(p_2^s / \text{kPa}) = 7.11572 - \frac{1684.123}{T/K - 43.568} \quad (18)$$

[19]

The $\bar{v}_i^L$ are determined from the modified Rackett equation as suggested by Spencer and Adler [29]:

$$\bar{v}_i^L = \frac{\bar{R}T_{crit}}{p_{crit}} z_{RA}^{1+\left(1-\frac{T}{T_{crit}}\right)^{2/7}} \quad (19)$$

where z_{RA} is the constant of the modified Rackett equation. For methanol, z_{RA} is given with 0.2323. The value for water was not reported. We thus estimate the value to 0.229 using the critical values as proposed by Spencer and Danner [30] when no data are available. The critical values were taken from the National Institute of Standards and Technology (NIST) for methanol [31] and water [32]. These values can be found in Table 1 [29].

For the pure components, the equation for the fugacity coefficient in the vapor phase is defined as:

Table 1

Used parameter for the Equilibrium method.

	Methanol	Water
T_{crit} (K)	513	647
p_{crit} (bar)	81	220.64
$\bar{v}_{crit}$ (m ³ /mol)	0.001/0.117	0.001/17.9
z_{RA}	0.2323	0.229
ω	0.572	0.344
a	0.0878	0.0279
b	0.056	0.0229

$$\varphi_i^S = \exp\left\{\frac{pB(T)}{RT}\right\} \quad (20)$$

and for the fugacity coefficients in the mixture:

$$\varphi_1 = \exp\left\{\frac{p}{RT}(2\bar{y}_1B_{11} + 2\bar{y}_2B_{12} - B_M)\right\} \quad (21)$$

$$\varphi_2 = \exp\left\{\frac{p}{RT}(2\bar{y}_1B_{12} + 2\bar{y}_2B_{22} - B_M)\right\} \quad (22)$$

Detailed derivations of the fugacity coefficients are provided in [appendix](#). All method specific component parameter can be found in [Table 1](#) [1,21].

The Equilibrium method can be directly applied to all other constant temperature and constant pressure data sets. For constant pressure cases, the second virial coefficients have to be calculated for each measured composition because the temperature changes between them. For constant temperature data sets B_{ii} and B_{ij} have to be calculated only once. To prove the correctness of the Equilibrium method, one recalculates the activity coefficients based on the equilibrium data from Kurihara et al. [19]. In [Fig. 3\(d\) - \(f\)](#), we display the results of the three investigated temperatures from these publications. The recalculated activity coefficients for water are always greater than those from Kurihara et al. The recalculated values for methanol have the opposite tendency and they are only slightly lower. The reason for the differences lies in the unknown parameters in Kurihara et al. which have to be taken from other cited publications or further references.

4.4. Wilson method based on Soujanya et al

Soujanya et al. [26] used the Wilson method to calculate the activity coefficients for methanol (1) and water (2) and out from them the vapor mole fractions. The activity coefficients are calculated as follow:

$$\ln\gamma_1 = -\ln(\bar{x}_1 + \bar{x}_2\Lambda_{12}) + \bar{x}_2\left(\frac{\Lambda_{12}}{\bar{x}_1 + \bar{x}_2\Lambda_{12}} - \frac{\Lambda_{21}}{\bar{x}_2 + \bar{x}_1\Lambda_{21}}\right) \quad (23)$$

$$\ln\gamma_2 = -\ln(\bar{x}_2 + \bar{x}_1\Lambda_{21}) - \bar{x}_1\left(\frac{\Lambda_{12}}{\bar{x}_1 + \bar{x}_2\Lambda_{12}} - \frac{\Lambda_{21}}{\bar{x}_2 + \bar{x}_1\Lambda_{21}}\right) \quad (24)$$

The Wilson parameters Λ_{12} and Λ_{21} are obtained for each constant pressure set. Soujanya et al. [26] minimized the objective function F :

$$F = \left[\frac{P_{exp}}{\gamma_1\bar{x}_1P_1^S + \gamma_2\bar{x}_2P_2^S} - 1\right]^2 \quad (25)$$

Here the activity coefficients from [Eqs. \(23\) and \(24\)](#) are inserted in the function. p_{exp} is the measured equilibrium pressure. The p_i^S are obtained from the Antoine equation in the following forms:

$$\ln\left(\frac{p_1^S}{0.133 \text{ kPa}}\right) = 18.5875 - \frac{3626.55}{T/K - 34.29} \quad (26)$$

$$\ln\left(\frac{p_2^S}{0.133 \text{ kPa}}\right) = 18.3036 - \frac{3816.44}{T/K - 46.13} \quad (27)$$

[26]

To apply the Wilson method to other experimental data sets, we initially recalculate the Wilson parameters using the same optimization scheme, thereby proving the accuracy of our recalculated ones. [Table 2](#) contrasts the obtained Wilson parameter with the Wilson parameter from Soujanya et al. [26]. The Wilson parameters are very similar apart from the 0.2928 bar case. No clear tendency is observed in the activity coefficients from Soujanya et al. for six different pressures. Especially at 0.2936 bar, the activity coefficient for infinity diluted methanol is 2.46, which is much greater than that for all following greater pressures. The recalculation of the activity coefficient with the Wilson parameter from Soujanya et al. itself give much lower values ([Fig. 4\(b\)](#) black empty squares) than ones written in the publication ([Fig. 4\(b\)](#) black filled circles). This is more reasonable because it follows a trend in the activity coefficients among all six pressures.

The Wilson method was adopted to other experimental data sets with constant pressure as described before. For data sets with constant system temperatures the optimization method needed a different way of calculation. To keep as much as possible from the original constant pressure method first the Wilson parameter were calculated for each mole fraction which represents a different pressure under the same temperature. Afterwards an additional optimization was done over all so obtained Wilson parameter to have only one set of them for each constant temperature data set. All Wilson parameter for the different constant pressure and constant temperature cases are listed in [Tables 3 and 4](#). [Fig. 4\(a\) - \(f\)](#) show the good agreements between the activity coefficients calculated from Soujanya et al. [26] and our recalculated ones (grey circles). The different Wilson parameters for 0.2938 bar ([Table 2](#)) implies their influence on the activity coefficients in [Fig. 4\(b\)](#) particularly at higher mole fractions of each component, where our Wilson parameters yield slightly greater activity coefficients.

4.5. Thermodynamic consistency

To check the thermodynamic consistency of the experimental and calculated data three different test are used.

4.5.1. Area test

The quality of the measured VLE data is evaluated with the so-called area test, as suggested by Redlich and Kister in 1948 [33]. For isothermal data sets, the following condition needs to be satisfied:

$$\int_{\bar{x}_1=0}^{\bar{x}_1=1} \ln\left(\frac{\gamma_1}{\gamma_2}\right) d\bar{x}_1 = 0 \quad (28)$$

The activity coefficients obtained in this work, either from publications or calculations, can be plotted in a $\bar{x}_1 - \ln(\gamma_1/\gamma_2)$ - diagram. A second-order polynomial function is fitted to all data points. The integral between $\bar{x}_1 = 0$ and the zero point of the polynomial function yields the area A_1 . The area A_2 is obtained similarly by integrating between the zero point and $\bar{x}_1 = 1$. Thermodynamic consistency is achieved when the criterion is met:

$$D = \frac{||A_1| - |A_2||}{|A_1| + |A_2|} 100 < 10 \% \quad (29)$$

Table 2

Wilson parameter from optimization at different pressures from Soujanya et al. [26] and recalculated in these work.

p (bar)	Λ_{12} (Soujanya et al.)	Λ_{21} (Soujanya et al.)	Λ_{12}	Λ_{21}
0.1519	0.9045	0.6266	0.904132	0.626234
0.2938	0.8991	0.6036	0.414521	0.96871
0.4266	0.8919	0.5799	0.89229	0.578325
0.5603	0.8957	0.5963	0.896928	0.593905
0.6738	0.8851	0.5441	0.877006	0.550264
0.9530	0.8941	0.6116	0.881675	0.628766

Table 3

Wilson parameter for constant pressure data sets.

Source	p (bar)	Λ_{12}	Λ_{21}
Gao et al. [18]	1.01325	0.34524222	1.15699748
Kato et al. [28]	1.01325	0.3834422	1.05824397

Table 4

Wilson parameter for constant temperature data sets.

Source	T (K)	Λ_{12}	Λ_{21}
Kurihara et al. [19]	323.15	0.296864	1.261253
	328.15	0.36654	1.171089
	333.15	0.389893	1.136751
Poling et al. [22]	298.15	0.794741	0.754291
	323.15	0.467443	1.039421
	373.15	0.206676	1.595843
	318.15	0.549808	0.904498
McGlashan and Williamson [24]	308.15	0.255292	1.381556
	323.15	0.467168	1.039785
	338.15	0.409544	1.092349
	323.15	0.467168	1.039785

The deviation D considers the experimental error, the extrapolation error of the polynomial fitting function and the integration error [1,14]

The deviation for constant temperature data sets can be found in Table 5. As we used three different methods to calculate the activity coefficients from the experimental equilibrium data, the deviation is calculated for each method. Some experimental data do not meet the thermodynamic consistency criteria under the Ideal Equilibrium method and the Equilibrium method. The activity coefficients from the Wilson model always satisfies the thermodynamic consistency. For 232.15 K, 338.15 K, and 373.15 K, only activity coefficients obtained from Wilson are appropriate to compare with the different UNIQUAC and the UNIFAC model.

For isobaric conditions, Eq. (28) cannot be directly applied. Another term accounting for the heat effect of mixing has to be included. Since the experimental data related to the term is difficult to measure, Wisniak et al. [14] concluded that Eq. (29) is also valid for isobaric conditions. We list the deviations computed with Eq. (29) in Table 6. We observe that most deviations are much lower than 10 %, and the only value exceeding the limit appears for the Equilibrium method using the experimental data from Gao et al.

4.5.2. Point test

According to the Gibbs phase rule, the measured vapor mole fractions represent an additional information that can be used to check the thermodynamic consistency. Fredenslund et al. [34] suggested the following condition for the thermodynamic consistency: [14]

$$\sum_{i=1}^n \frac{|\bar{y}_{i,exp} - \bar{y}_{i,calc}|}{n} \leq 0.01 \quad (30)$$

Tables 7 and 8 show the differences between the measured and

Table 5Deviation D in % using the activity coefficient from the source, the Ideal Equilibrium model, the Equilibrium model and the Wilson model for constant temperature data.

Source	Temperature in K	Source	Ideal Equilibrium	Equilibrium	Wilson
Kurihara et al. [19]	323.15	7.73	3.72	0.63	4.15
	328.15	8.58	4.06	0.79	0.63
	333.15	6.06	0.97	5.27	0.08
Poling et al. [22]	298.15	-	7.35	3.95	0.005
	323.15	-	13.93	17.81	0.09
	373.15	-	25.96	31.56	0.47
	318.15	-	0.53	3.51	0.16
Yao et al. [23]	308.15	-	3.19	6.05	1.19
	323.15	-	16.5	20.37	0.12
	338.15	-	12.84	17.82	0.06
	323.15	-	12.84	17.82	0.06

Table 6Deviation D in % using the activity coefficient from the source, the Ideal Equilibrium model, the Equilibrium model and the Wilson model for constant pressure data.

Source	Pressure in bar	Source	Ideal Equilibrium	Equilibrium	Wilson
Soujanya et al. [26]	0.1519	0.02	0.42	2.36	0.02
	0.2938	0.07	0.02	3.09	0.16
	0.4266	0.03	0.08	4.51	0.04
	0.5603	0.03	0.15	5.58	0.03
	0.6738	0.05	0.11	5.44	0.03
Gao et al. [18]	0.9530	0.03	0.21	7.25	0.02
	1.01325	0.29	8.19	12.78	2.07
Kato et al. [28]	1.01325	1.12	1.55	7.37	0.89

experimental vapor mole fraction and pressures. The thermodynamic consistency is fulfilled for most of the results. Similarly to the area test, the 373.15 K case does not reach the thermodynamic consistency. Additionally, the 308.15 K case slightly exceeds the condition, which can be induced by a single outlier in the experimental data. For the constant pressure cases, only the 0.953 bar case is slightly over the condition.

4.5.3. Gibbs energy of mixing

To examine whether the temperature dependency of the interaction parameter for isobaric equilibrium data and the pressure dependency of the interaction parameter for isothermal equilibrium data are properly considered, a graphical test with the dimensionless Gibbs energy of mixing ($\bar{g}/R/T$) was introduced by Marcilla et al. [35,36]. The dimensionless Gibbs energy of mixing is calculated for the liquid and the vapor phases based on the liquid mole fractions and the corresponding equilibrium vapor mole fractions and temperatures for isobaric conditions or the pressures for isothermal conditions. The required saturation vapor pressures are obtained with Eq. (3) and Eq. (4). A fourth-order polynomial function is fitted to all obtained points for the Gibbs energy of mixing of the liquid phase. Each pair of the Gibbs energy of mixing for the liquid and the vapor phases is connected by a line. The line should be a tangent to the fitted curve of the Gibbs energy of mixing for the liquid phase.

The test is performed for all isobaric and isothermal experimental data as well as for the UNIQUAC TPD V2 model and the UNIQUAC Islam V2 model. Fig. 1 displays the results for the 1.01325 bar case and the 323.15 K case. All other graphical results can be found in the supplementary material. All figures show that the lines are tangents to the fitted curve of the Gibbs energy of mixing for the liquid phase. That implies that the pressure and the temperature dependencies are chosen correctly.

Table 7Deviation between experimental and calculated vapor mole fraction $\Delta\bar{y}_1^a$ and pressure Δp^b for constant temperatures.

Source	T (K)	UNIQUAC TPD V2		UNIQUAC Islam V2		UNIQUAC TD V2	
		$\Delta\bar{y}_1$	$\Delta p/\text{Pa}$	$\Delta\bar{y}_1$	$\Delta p/\text{Pa}$	$\Delta\bar{y}_1$	$\Delta p/\text{Pa}$
Kurihara et al. [19]	323.15	0.005	470.645	0.005	330.191	0.005	325.142
	328.15	0.004	712.175	0.004	524.460	0.004	516.396
	333.15	0.003	808.506	0.002	630.302	0.003	675.153
Poling et al. [22]	298.15	0.007	120.340	0.005	83.498	0.008	121.705
	323.15	0.003	352.381	0.004	251.877	0.006	220.673
	373.15	0.016	17,370.931	0.021	10,377.500	0.016	6958.992
Yao et al. [23]	318.15	0.005	243.459	0.005	228.344	0.004	227.583
McGlashan and Williamson [24]	308.15	0.012	323.963	0.011	315.840	0.012	209.288
	323.15	0.005	362.949	0.006	275.611	0.008	272.994
	338.15	0.005	592.959	0.004	535.716	0.007	484.450

$$^a \Delta\bar{y}_1 = \left(\sum |\bar{y}_{1,exp} - \bar{y}_{1,calc}| \right) / N,$$

$$^b \Delta p = \left(\sum |p_{exp} - p_{calc}| \right) / N \text{ with } N \text{ as number of experimental data points.}$$

Table 8Deviation between experimental and calculated vapor mole fraction $\Delta\bar{y}_1^a$ and pressure ΔT^b for constant pressures.

Source	p (bar)	UNIQUAC TPD V2		UNIQUAC Islam V2		UNIQUAC TD V2	
		$\Delta\bar{y}_1$	$\Delta T/K$	$\Delta\bar{y}_1$	$\Delta T/K$	$\Delta\bar{y}_1$	$\Delta T/K$
Soujanya et al. [26]	0.1519	0.005	0.239	0.006	0.245	0.006	0.313
		*					
	0.2938	0.009	0.567	0.008	0.558	0.008	0.557
		*					
	0.4266	0.007	0.324	0.007	0.369	0.011	0.415
		*					
	0.5603	0.008	0.390	0.012	0.410	0.009	0.445
		*					
	0.6738	0.008	0.407	0.008	0.431	0.009	0.468
		*					
	0.9530	0.012	0.573	0.012	0.600	0.011	0.564
		*					
Gao et al. [18]	1.01325	0.006	0.374	0.005	0.337	0.006	0.353
		*					
Kato et al. [28]	1.01325	0.006	0.131	0.005	0.138	0.007	0.117

$$^a \Delta\bar{y}_1 = \left(\sum |\bar{y}_{1,exp} - \bar{y}_{1,calc}| \right) / N,$$

$$^b \Delta T = \left(\sum |T_{exp} - T_{calc}| \right) / N \text{ with } N \text{ as number of experimental data points.}$$

* $\bar{y}_{1,exp}$ calculated value from the source as it was not measured.

5. Results and discussion

We firstly compare the calculated activity coefficients computed by the three methods Ideal Equilibrium, Equilibrium and Wilson. For the 1.01325 bar case, activity coefficients from two experimental data sets are available. Looking first at methanol in Fig. 4(g). Compared to the data from Kato et al. [28], the results for the activity coefficients obtained using the three methods are very similar, as displayed in Fig. 4(g) for methanol. For Gao et al. [18], both Ideal Equilibrium and Equilibrium are lower at larger methanol mole fractions. The reason is demonstrated in the right panel of Fig. 2 where there is a slight difference between the two data sets from Gao et al. and Kato et al. in the temperature-composition diagram. The temperature differences can lead to variations in saturation vapor pressures as well as corresponding activity coefficients. Gao et al. calculated the vapor mole fraction whereas Kato et al. measured the vapor mole fraction which leads to further differences that can be visible in the composition diagram Fig. 2 (left). The experimental values from Kato et al. are expected to be more reliable than ones from Gao et al. as the activity coefficients for pure methanol are lower than one, in Gao et al. case. In principle, the activity coefficient for pure components is one. In addition, for water at high water mole fractions, the calculated activity coefficients with the data from Gao et al. are lower than the values from the publication. The trend changes at lower water mole fractions. Here, the activity coefficients are greater and these for both experimental data sets. The Equilibrium method tends to yield higher activity coefficients whereas using the Ideal Equilibrium method, the activity coefficients are closer to the

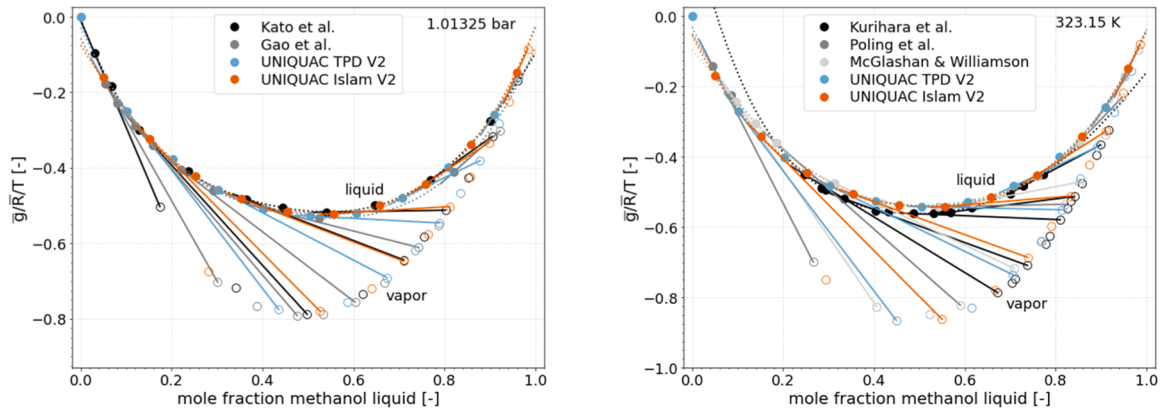


Fig. 1. Dimensionless Gibbs energy of mixing for the liquid phase (filled symbols) and the vapor phase (not filled symbols) over the liquid mole fraction of methanol for 1.01325 bar and 323.15 K. Dashed line represents the fitted curve of the Gibbs energy of mixing for the liquid phase. Solid lines connect corresponding Gibbs energy of mixing for the liquid and the vapor phase. Experimental data from Kato et al. [28], Gao et al. [18], Kurihara et al. [19], Poling et al. [22] and McGlashan and Williamson [24].

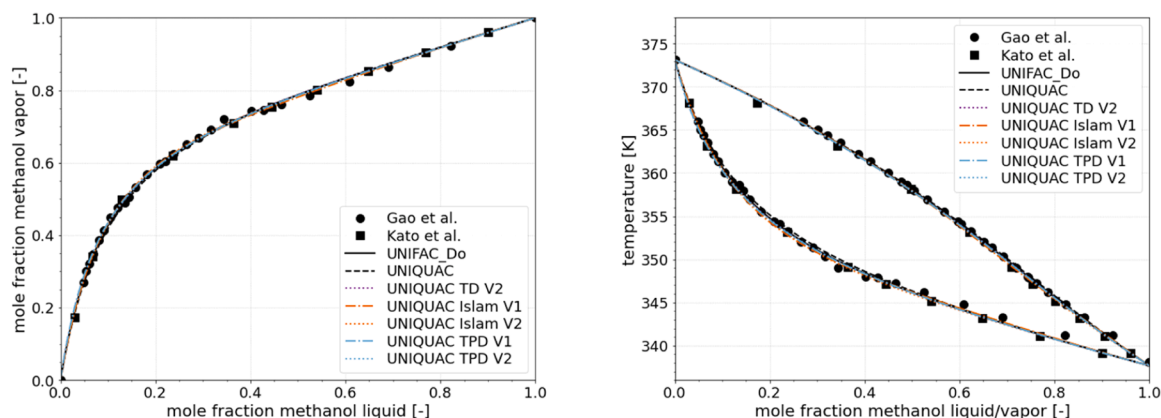


Fig. 2. Composition diagram (left) and temperature-composition diagram (right) for methanol-water mixtures at 1.01325 bar. Experimental data from Gao et al. [18] and Kato et al. [28].

values in the publications. Overall, the activity coefficients obtained by the Wilson method are very similar to those reported by Gao et al. and Kato et al., who also employed the same method. At lower water mole fractions, the calculated activity coefficients exhibit a bigger spread among the different methods.

For pressures below 1 bar (see Fig. 4(a) - (f)), all three methods predict nearly identical activity coefficients, except for water at lower water mole fractions, where the Equilibrium method shows higher activity coefficients than other methods. In fact, at low pressures, the vapor can be treated as an ideal gas, which describe why our considered methods give the same results.

In Fig. 3(a) - (h), the variation in the activity coefficient among the three methods is observed for the constant temperature cases, where the activity coefficients for methanol also show greater variations. Interestingly, we find that the largest differences appear at the two highest temperature cases, and this is more pronounced for water.

The three methods Ideal Equilibrium, Equilibrium and Wilson show a stronger dependency on temperature than on pressure. The empty symbols in Figs. 3 and 4 represent the activity coefficients calculated with the 3 methods based experimental data. The filled symbols are the activity coefficients directly taken from the publications. The simulated UNIFAC Do and the different UNIQUAC activity coefficients should be within the region indicated by the symbols to show the correct reproduction of temperature and pressure dependency.

We simulate the activity coefficients for the experimental conditions using UNIFAC Do, UNIQUAC, UNIQUAC Islam, UNIQUAC TD and UNIQUAC TPD. The parameters for UNIQUAC Islam, UNIQUAC TD and UNIQUAC TPD are obtained with the previously described methods (see 3.1 UNIQUAC TD, 3.2 UNIQUAC Islam and 3.3 UNIQUAC TPD). For the first version (V1), all published activity coefficients are used as reference values (see 4.1 Experimental Data), excluding the 0.2936 bar data from Soujanya et al. [26]. The data are in a pressure range of 0.1519 bar to 1.01325 bar and in a temperature range of 323.15 K and 372.55 K. A second optimization (V2) was carried out, containing all activity coefficients from the named literature excluding the 0.2936 bar data from Soujanya et al. [26] as well. We further added the activity coefficients obtained by the Equilibrium method without considering four outliers for all datasets from Poling et al. [22], McGlashan and Williamson [24] and Yao et al. [23]. The temperature range thus was extended to 298.15 K - 373.15 K. The obtained parameters for V1 and V2 are listed in Tables 9, 10 and 11.

5.1. Constant temperature

Simulations were performed for six different temperatures between 298.15 K and 373.15 K (see Fig. 3(a) - (h)). For UNIQUAC Islam V1, the activity coefficients for methanol show a peak at lower temperatures.

The trend is also visible until temperatures of 323.15 K. For UNIQUAC TPD V1, the activity coefficients of water at high methanol concentration and lower temperatures tend to increase rapidly. For 373.15 K, both methods exhibit a weird behaviour. This is mainly due to a lack of optimization points at both low and high temperatures.

The second optimization version (V2), in turn, was applied to overcome this problem. Now, the trends of the activity coefficients of UNIQUAC Islam V2 and UNIQUAC TPD V2 fit much better to the triangles calculated from experimental values. For instance, at 298.15 K, the activity coefficients for both V2 methods for water lie closely on each other in the region of the experimentally calculated values, as shown in Fig. 3 (a). The UNIQUAC TPD V2 predicts higher activity coefficients than the UNIQUAC Islam V2. Both fit to the experimentally calculated triangles. The UNIQUAC TD V2 model predicts lower activity coefficients for water than the UNIQUAC TPD V2 model. The activity coefficients for water predicted by UNIFAC Do, which also has a temperature dependency, are also within the range of the experimental data.

In the case of methanol, UNIQUAC Islam V2 and UNIQUAC TPD V2 are quite similar. On the other hand, at lower temperatures, the deviation is the greatest (Fig. 3(a)-(b)). For low methanol mole fractions, the activity coefficients computed by UNIQUAC TPD V2 are higher than those by UNIQUAC Islam V2. UNIQUAC Islam V2 fits generally better than UNIQUAC TPD V2. We observe that the deviation from the experimental data indicated by triangles is the greatest at higher mole fractions for 373.15 K (Fig. 3(h)). The lower seven temperatures are within 40 K whereas 373.15 K is 35 K higher than the next lower temperature. Additional data points at higher temperatures therefore are required in the optimization process to achieve better performance. The UNIQUAC TD model predicts higher activity coefficients for methanol at low methanol mole fractions compared to the UNIQUAC TPD model. Its prediction tends to be closer to those of the UNIFAC Do, though still slightly lower. Both UNIQUAC TD and UNIFAC Do include only temperature dependency. UNIFAC Do predicts greater activity coefficients for methanol than UNIQUAC Islam V2 and UNIQUAC TPD V2. The UNIFAC Do treats mixtures as combinations of functional groups. Methanol and water are functional groups themselves. Therefore, the interaction parameters could be directly obtained from experimental data of mixtures of methanol and water at different temperatures. This explains the good agreement to the experimental activity coefficients. The UNIQUAC TPD V2 was optimized in parallel with the pressure as well. UNIQUAC, which has no temperature dependent binary interaction parameter, fits very well with the results from UNIFAC Do for temperatures between 318.15 K and 338.15 K which is a limited temperature range of 20 K. For lower or higher temperatures, the differences increase. As expected, the UNIQUAC is less flexible.

Table 7 shows the deviations of the experimental and calculated vapor mole fractions and pressures for the three model UNIQUAC TD V2,

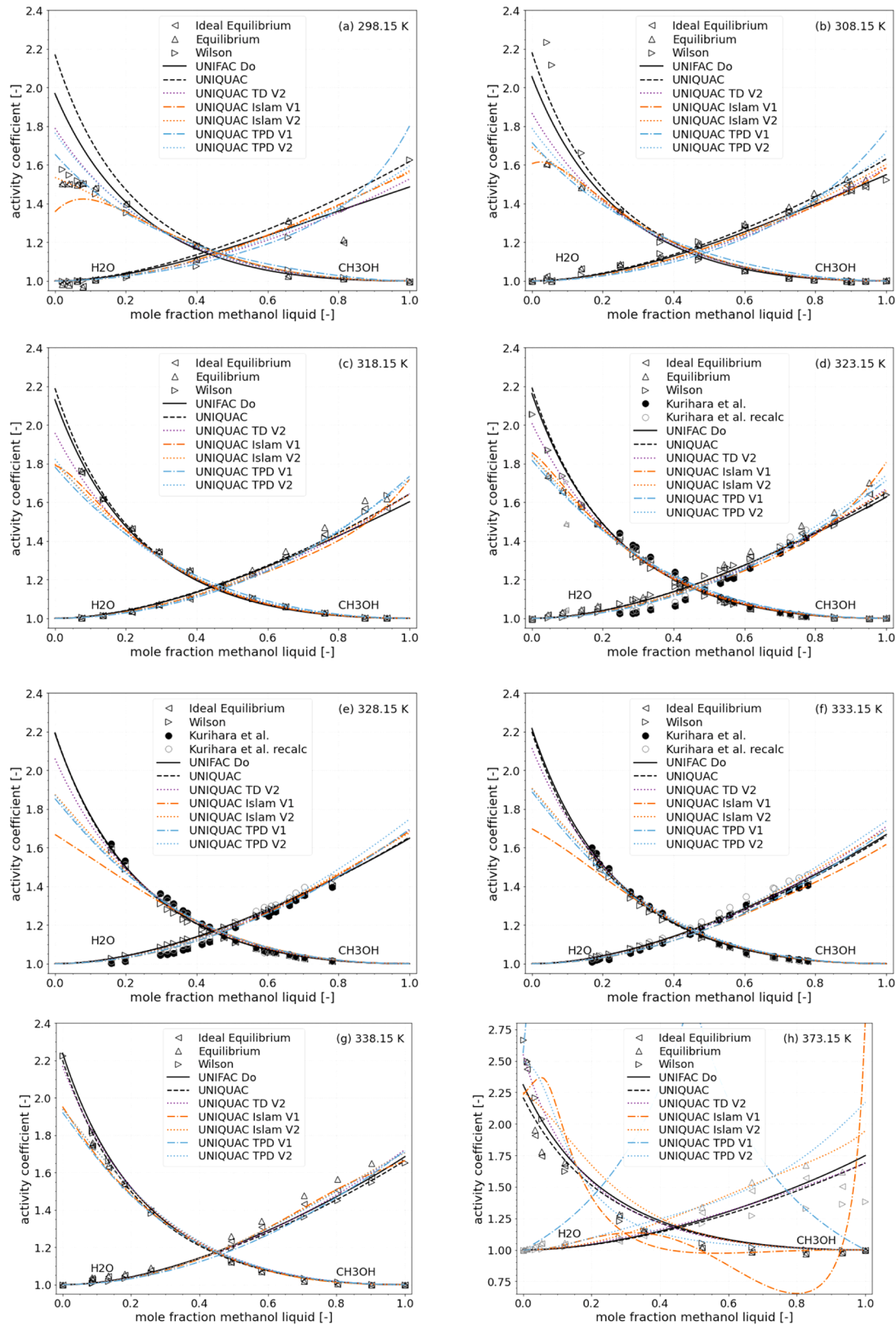


Fig. 3. Activity coefficients over liquid mole fraction of methanol for the binary methanol-water mixtures at (a) 298.15 K, (b) 308.15 K, (c) 318.15 K, (d) 323.15 K, (e) 328.15 K, (f) 333.15 K, (g) 338.15 K, (h) 373.15 K. (a) triangles based on data from Poling et al. [22], (b) triangles based on data from McGlashan and Williamson [24], (c) triangles based on data from Yao et al. [23], (d) grey triangle based on data from Poling et al. [22] and black triangle based on data from McGlashan and Williamson [24] and Kurihara et al. [19], (e)-(f) triangles based on data from Kurihara et al. [19], (g) triangles based on data from McGlashan and Williamson [24], (h) triangles based on data from Poling et al. [22].

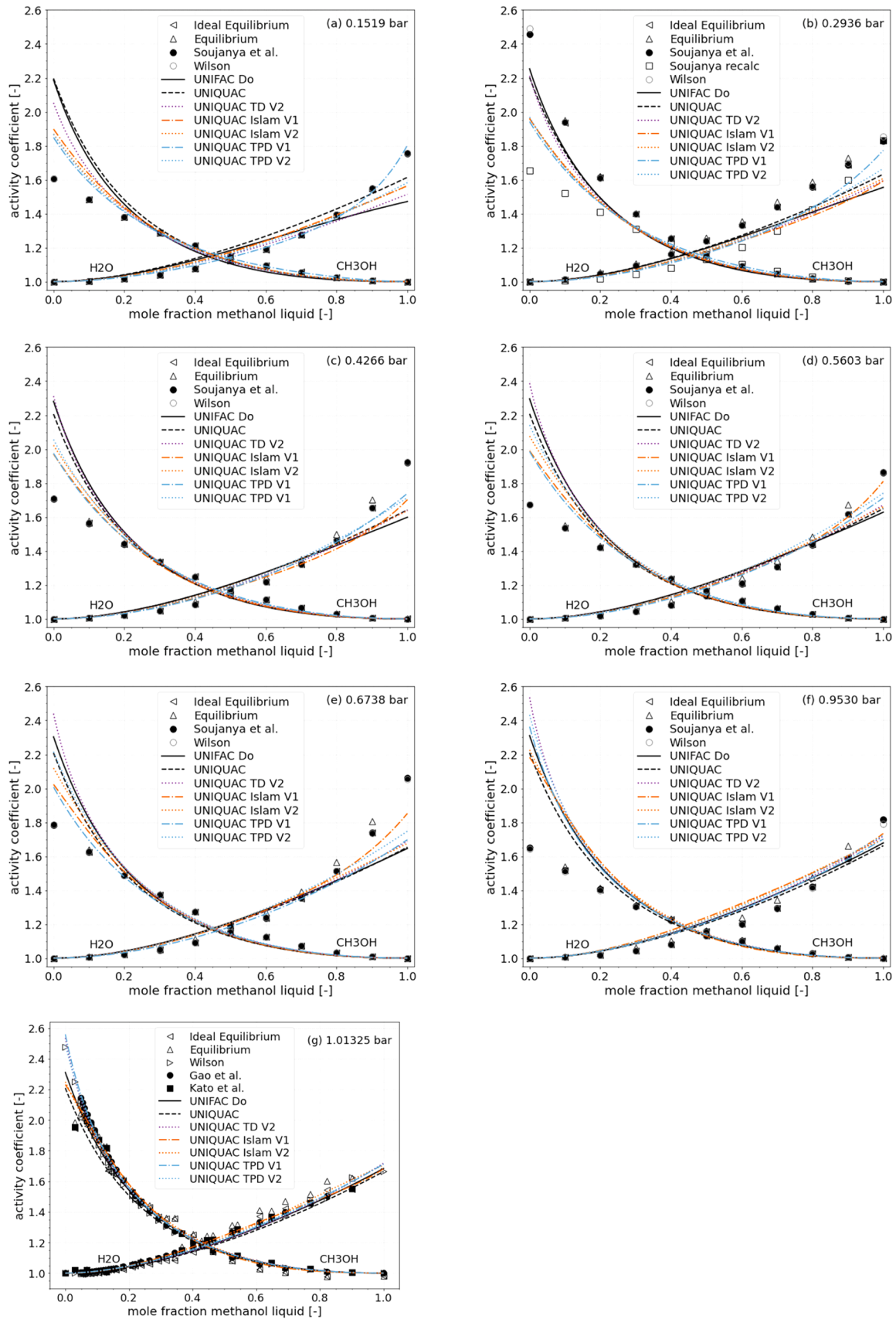


Fig. 4. Activity coefficients over liquid mole fraction of methanol for the binary methanol-water mixtures at (a) 0.1519 bar, (b) 0.2936 bar, (c) 0.4266, (d) 0.5603 bar, (e) 0.6738, (f) 0.9530 bar, (g) 1.01325 bar. (a)-(f) triangles based on data from Soujanya et al. [26], (g) triangles based on data from Gao et al. [18] and Kato et al. [28].

Table 9

Optimized parameter for UNIQUAC Islam (Eq. (9)) for two different data sets.

	UNIQUAC Islam V1		UNIQUAC Islam V2	
	Δu_{12}	Δu_{21}	Δu_{12}	Δu_{21}
A_{ij}	-1086.12257	135.641054	-186.401647	111.484145
B_{ij}	2131902.67	-314252.477	5.03975689	-1.24858385
b_{ij}	21.7909894	18.9892547	-5.02162647	-6.73928672
C_{ij}	1485.39653	1245743.02	94028.7253	512.867447
c_{ij}	0.0513124267	1.41087608	1.87434473	0.338726523
K_{ij}	-77560.1671	-195101.859	-12713.9189	27390.0009

Table 10

Optimized parameter for UNIQUAC TPD (Eq. (12)) for two different data sets.

	UNIQUAC TPD V1		UNIQUAC TPD V2	
	Δu_{12}	Δu_{21}	Δu_{12}	Δu_{21}
A_{ij}	454.861292	-259.838572	76.0813178	-79.4065590
B_{ij}	-2.70366053	1.00003426	-0.895562221	1.01333434
C_{ij}	2.68649979e-3	1.42443141e-3	4.72225167e-4	2.13275928e-4

Table 11

Optimized parameter for UNIQUAC TD (Eq. (7)).

	UNIQUAC TD V2	
	Δu_{12}	Δu_{21}
A_{ij}	-68.3941517	1496.05316
B_{ij}	2.60699317	-13.0992104
C_{ij}	-8.09695631e-3	0.0271577229

UNIQUAC TPD V2 and UNIQUAC Islam V2. A graphical representation in the composition diagrams and the pressure-composition diagrams can be found in the supplementary material. The highest deviation occurs for 308.15 K and 373.15 K. At 308.15 K, a greater spread between the activity coefficients predicted from the experimental data with the Wilson method and the Equilibrium methods is exhibited in Fig. 3(b).

5.2. Constant pressure

Most of the experimental constant pressure data used in this work are measured at around 1 bar. Only Soujanya et al. [26] published activity coefficients below 1 bar, but the tendency of the activity coefficients as pressure increase is unclear, therefore, achieving an accurate fit with it is difficult. The results can be seen in Fig. 4(a) - (g).

The activity coefficients obtained with the UNIQUAC produces almost same results for all pressures. The activity coefficients for methanol at infinite dilution is always 2.2, the activity coefficient for water at infinite dilution is changing slightly between 1.61 for 0.1519 bar to 1.66 at 1.01325 bar. The UNIFAC Do also changes a little the activity coefficients depending on pressure. As there is no direct pressure dependency in the model, the small changes arise from the calculation of Antoine equation with a given temperature.

In general, the activity coefficients are expected to increase with increasing pressure. Since the experimental data do not clearly follow this trend across all pressures, the optimized parameters may not reach their optimal performance. Both UNIQUAC TPD V2 and UNIQUAC Islam V2 show a similar behavior. UNIQUAC Islam V2 slightly underestimates activity coefficients at infinite dilutions, which are closer to

experimental data. Except for 1.01325 bar, all models are not able to accurately capture the experiment at lower mole fractions. Although the UNIQUAC Islam V2 method currently performs a little better than the UNIQUAC TPD V2, the UNIQUAC TPD V2 is not far behind in accuracy even with 6 parameters, which are half of the UNIQUAC Islam V2 method. Moreover, the method has the same quadratic form as temperature dependent UNIQUAC and UNIFAC Do. It is more applicable in the temperature dependent methods. The UNIQUAC TD V2 model predicts activity coefficients which are close to those calculated by the UNIFAC Do model, which is expected because it also includes only temperature dependency.

For the 1.01325 bar case, more data sets are available but the activity coefficients for infinite dilution are missing. Nevertheless, the trend can be followed quite well as many data are available between a methanol mole fraction of 0.1 to 0.9. Fig. 4(g) exhibits the good representation of all simulated activity coefficients. Fig. 2 illustrates this good agreements ones more in the composition diagram and the temperature-composition diagram. Diagrams for other pressures can be found in the supplementary material. We note that since the amount of experimental data used at constant pressure is quite limited, it is believed that the models could be further improved with the use of more data.

6. Conclusion

In summary, we suggested a new temperature and pressure dependent binary interaction coefficient for UNIQUAC. Overall, the version 2 (V2) parameters provide a better fit to the constant temperature and pressure experimental data compared to the version 1 (V1) parameters because more data were used in the optimization process. For this, we added the activity coefficients obtained with the Equilibrium methods based on experimental vapor-liquid equilibrium data. Despite using only 6 optimized parameter, UNIQUAC TPD V2 is comparable to UNIQUAC Islam V2, which has 12 optimized parameters. UNIQUAC Islam V2 is currently the better method to estimate the temperature and pressure dependent binary interaction parameter in UNIQUAC. Including a term where pressure and temperature are not together involved could make also UNIQUAC TPD V2 as sufficient as UNIQUAC Islam V2 or better. The UNIQUAC TPD V2 can be advantageous due to its simpler optimization and its possible applicability directly to UNIQUAC or UNIFAC.

CRedit authorship contribution statement

Adina Werner: Writing – original draft, Visualization, Software. **Jongmin Kim:** Writing – review & editing, Software. **Fabian Mauss:** Writing – review & editing, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.fluid.2025.114533](https://doi.org/10.1016/j.fluid.2025.114533).

Appendix. Calculation of Fugacity Coefficients

The calculation of the fugacity coefficients is done starting with the following equation:

$$\ln \varphi_i = \frac{1}{RT} \int_0^p \left[\left(\frac{\partial V}{\partial n_i} \right)_{T,p,n_j} - \frac{RT}{p} \right] dp \quad (\text{A.1})$$

The volume V in the partial derivation is replaced by the virial equation in Berlin form broken after the second term:

$$V = \frac{\bar{R}Tn_{\text{tot}}}{p} + n_{\text{tot}}B(T) \quad (\text{A.2})$$

with n_{tot} as total number of moles of the mixture and B as temperature dependent second virial coefficient. For the pure components in the vapor phase the equation for the fugacity coefficient is:

$$\varphi_i^S = \exp \left\{ \frac{pB(T)}{RT} \right\} \quad (20)$$

[21]

For the fugacity coefficients in the mixture the second virial coefficient has to be used as the one for the mixture:

$$B_M = \sum_i \sum_j \bar{y}_i \bar{y}_j B_{ij} \quad (\text{A.3})$$

The sum will include the second virial coefficients of the pure components (B_{11} , B_{22}) and the cross virial coefficients $B_{12} = B_{21}$. Using equation (A.3) for the partial derivation in equation (A.1) the equations for the virial coefficients of methanol (1) and water (2) in the mixture are:

$$\varphi_1 = \exp \left\{ \frac{p}{RT} (2\bar{y}_1 B_{11} + 2\bar{y}_2 B_{12} - B_M) \right\} \quad (21)$$

$$\varphi_2 = \exp \left\{ \frac{p}{RT} (2\bar{y}_1 B_{12} + 2\bar{y}_2 B_{22} - B_M) \right\} \quad (22)$$

[1,21]

The second virial coefficients are obtained after the method from Tsonopoulos [37]. For the pure components:

$$B_{ii} = \frac{\bar{R}T_{\text{crit}}}{p_{\text{crit}}} (b^{(0)}(T_R) + \omega b^{(1)}(T_R) + b^{(2)}(T_R)) \quad (\text{A.4})$$

with ω as acentric factor. The three dimensionless terms $b^{(0)}$, $b^{(1)}$ and $b^{(2)}$ are dependent on the reduced temperature $T_R = T/T_{\text{crit}}$. The term $b^{(2)}$ is added for polar species and is calculated over:

$$b^{(2)}(T_R) = \frac{a}{T_R^6} - \frac{b}{T_R^8} \quad (\text{A.5})$$

For the calculation of $b^{(0)}$ and $b^{(1)}$ refer to Tsonopoulos [37]. The cross coefficient B_{ij} , which is needed in the calculation of the B_M , can be calculated similarly to B_{ii} with equation (A.4). All component specific parameter (T_{crit} , p_{crit} , T_R , ω , a , b) have to be replaced by mixture specific ones.

Data availability

No data was used for the research described in the article.

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